



Marine Biology

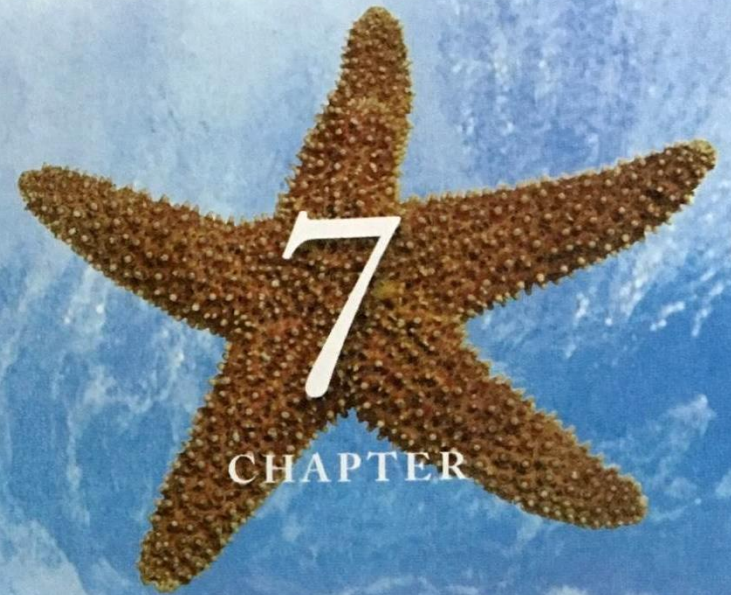
海洋生物学



Yanan Di (邸雅楠)

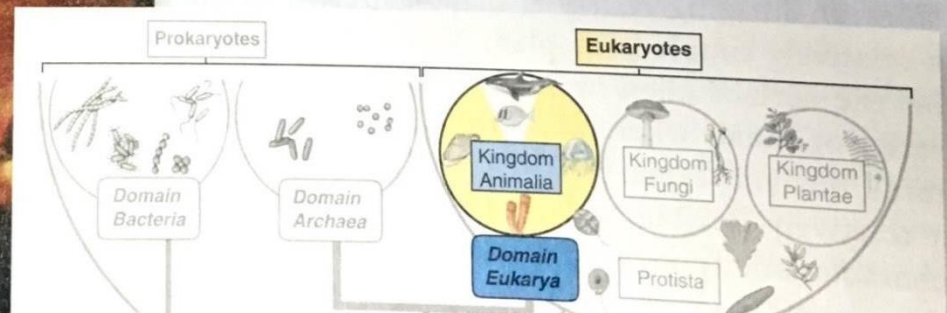
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CHAPTER

Marine Animals Without a Backbone

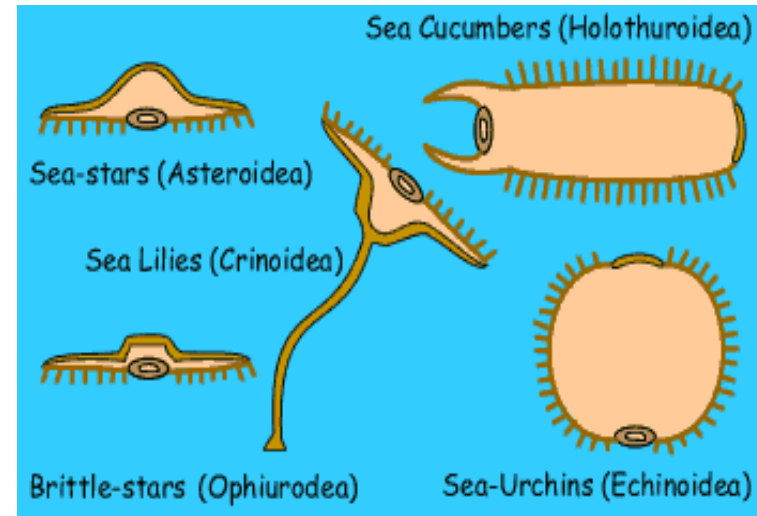


• Major Phyla of Marine Invertebrates

- | | |
|-----------------------|----------------------------|
| 1. Sponges (海绵动物) | 10. Mollusks (软体动物) |
| 2. Cnidarians (腔肠动物) | 11. Bryozoans (苔藓虫) |
| 3. Flatworms (扁形虫) | 12. Phoronids (帚虫) |
| 4. Ribbon worms (纽虫) | 13. Brachiopods (腕足类) |
| 5. Nematodes (线虫) | 14. Arrow worms (箭虫) |
| 6. Annelids (环节动物) | 15. Echinoderms (棘皮动物) |
| 7. Sipunculans (星虫动物) | 16. Hemichordates (半索海生动物) |
| 8. Echiurans (螠虫动物) | 17. Tunicates (被囊动物) |
| 9. Arthropods (节肢动物) | 18. Lancelets (文昌鱼) |

14. Echinoderms (棘皮动物)

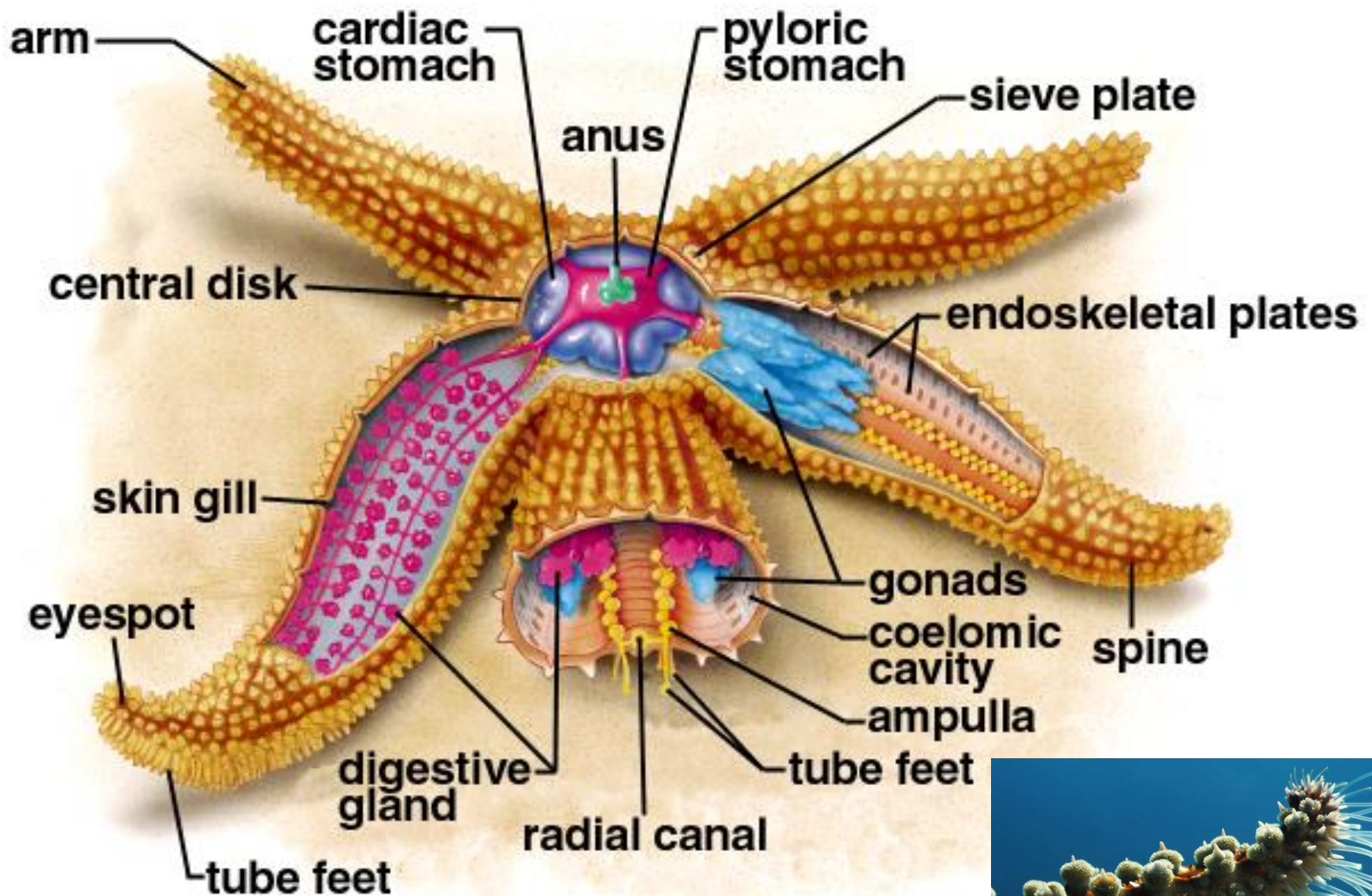
- Name means “Spiny Skin” , **Endoskeleton**, all 7000 species exclusively marine
- **Water vascular system with tube feet** important in feeding and locomotion
- **Skin gills** for respiration
- **Radial symmetry in adults, larvae are bilaterally symmetry**
- **Nervous system is decentralized**, no brain allows any portion of the body to lead
- Can regenerate lost body parts



14.1 Types of Echinoderms:

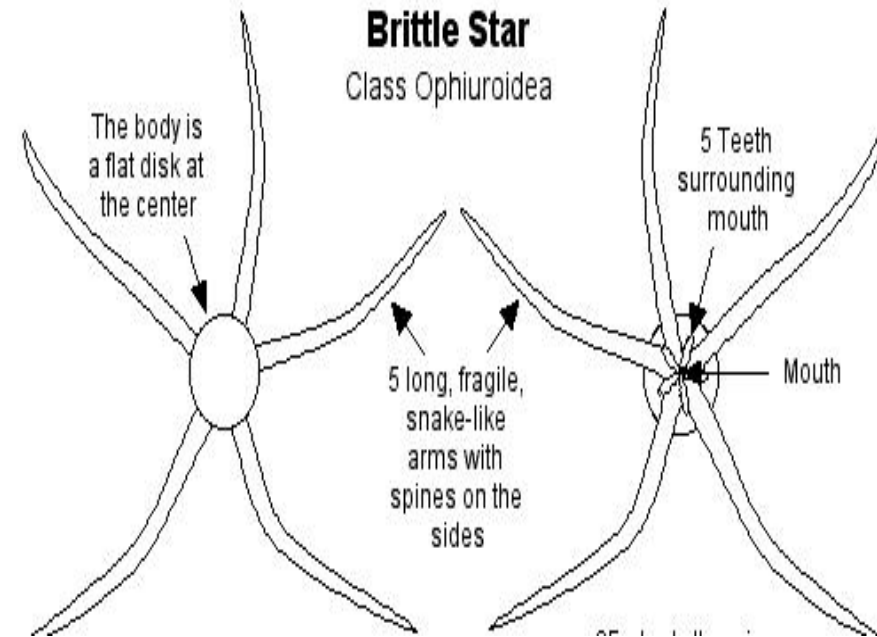
- Sea stars (Class of Asteroidea)
 - Move with tube feet
 - Have a **central disc** in center of body surrounded by five arms (or multiples of 5 arms – some species have 50 arms)
 - Internal organs extend through the entire body, including the arms
 - **Calcium carbonate plates** are loosely embedded in spiny skin making them slightly flexible
 - Carnivores that normally consume shellfish and coral





- Brittle stars (海蛇尾, Class of Ophiuroidea)

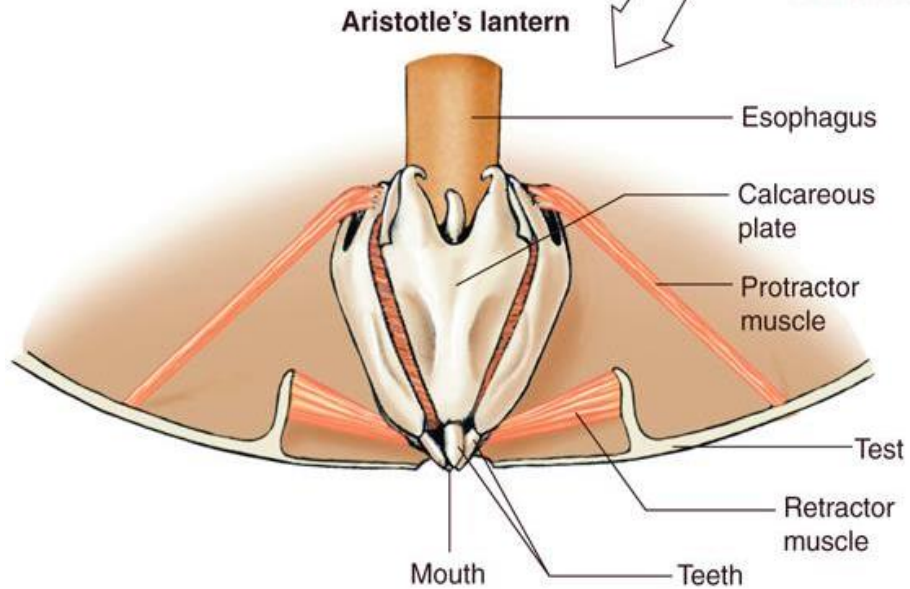
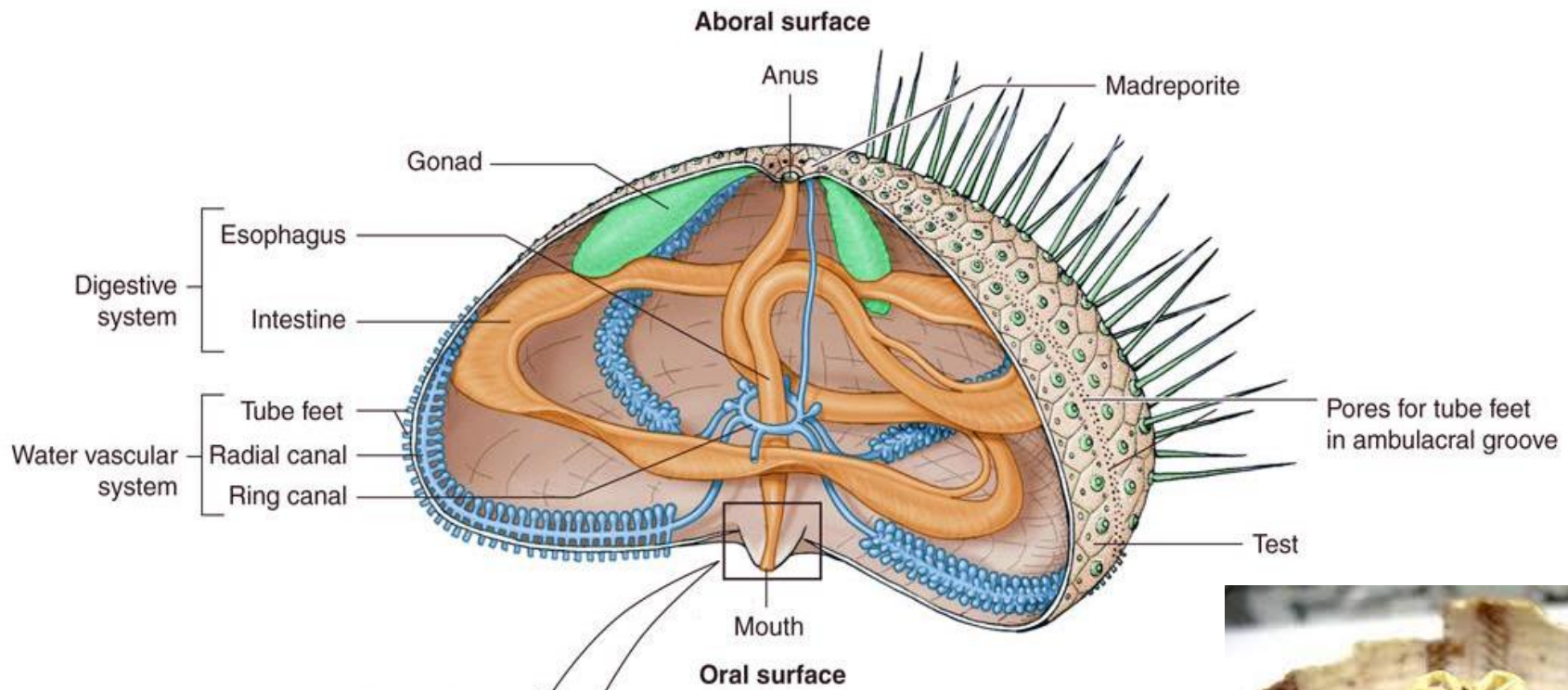
- Like sea stars, they have a **central disc** surrounded by arms
- **Five arms** seen in brittle stars are thin and covered in numerous spines
- **Internal organs are restricted to the central disc**
- The **tube feet** present in brittle stars are without suckers and used for feeding on detritus and small animals



- Sea Urchins/Sea Biscuits/Sand Dollars (Class of Echinoidea)

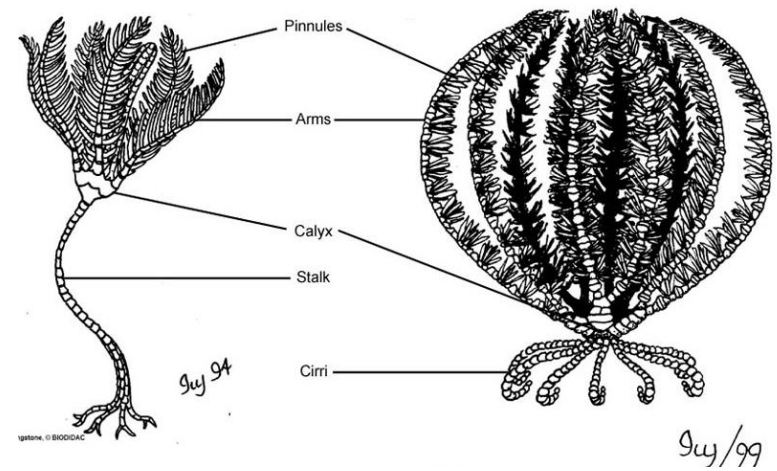
- Elongated, movable spines much longer than those in other groups
- Rigid plates are fused into a solid structure called a “**test**”
- Move with tube feet
- **Mouth on the bottom, anus on top of body**
- Biting mouth for grazing, **aristotles lantern** is the **feeding structure** of muscles and mouthpieces
- Feed on detritus, encrusting organisms, algae or anything else they can scrape off surfaces



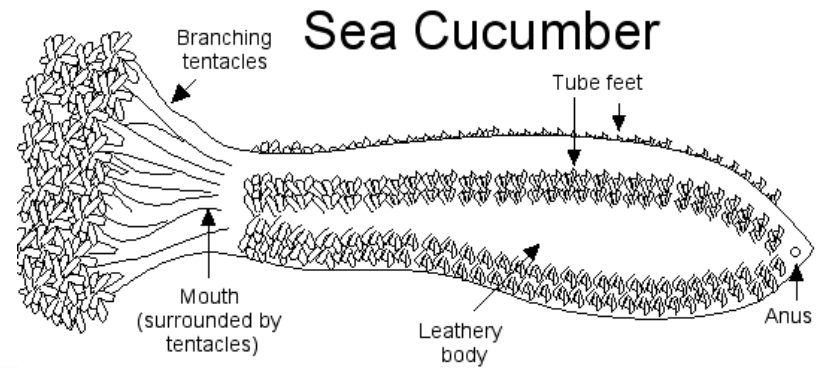
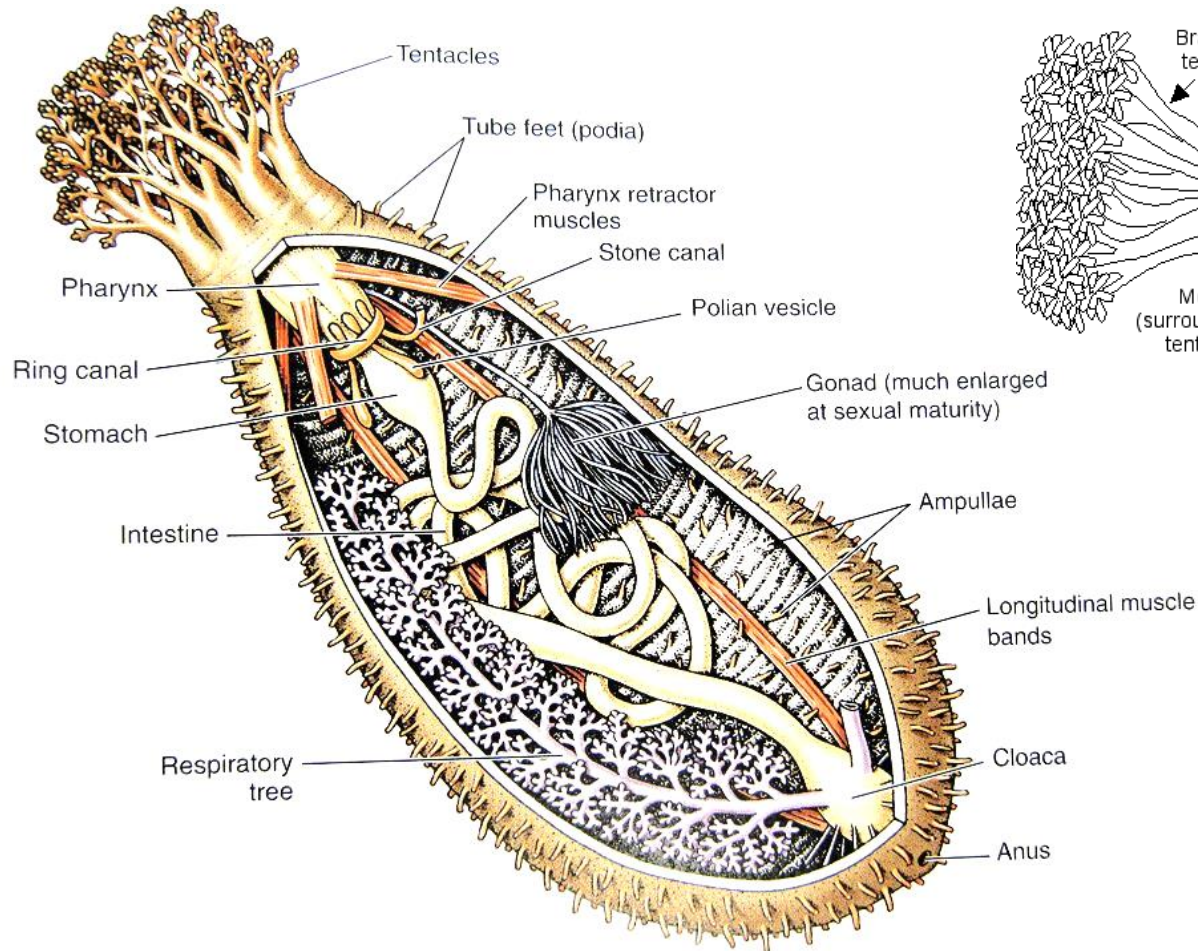


- Sea lilies (Class of Criniodea)

- Represented by feather stars and sea lilies
- 600 species are typically found in deep water
- Sea lilies live attached while feather stars are mobile
- These organisms have 5 or more arms that branch out for suspension feeding
- Some use a mucous net to aid in food capture



- Sea cucumbers (海参, Class of Holothuroidea)
 - 5 rows of 2 feet are restricted to one side, where the animal lies
 - The plates found in the sea cucumbers are loosely embedded in the thick skin



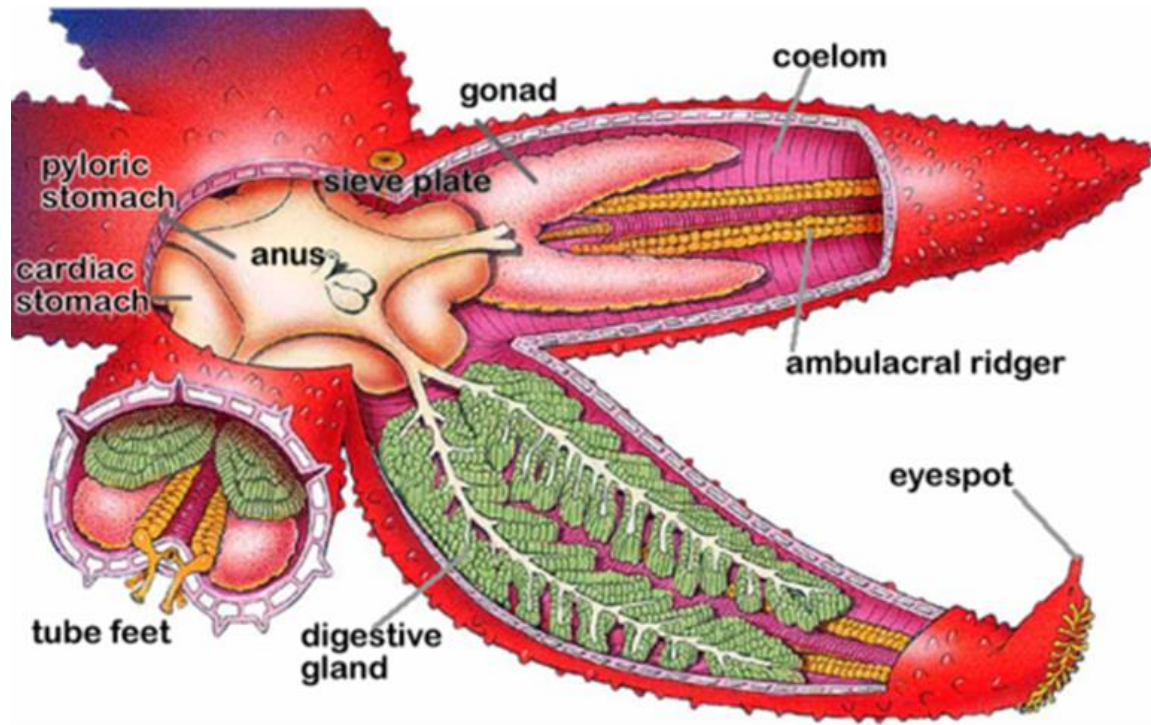
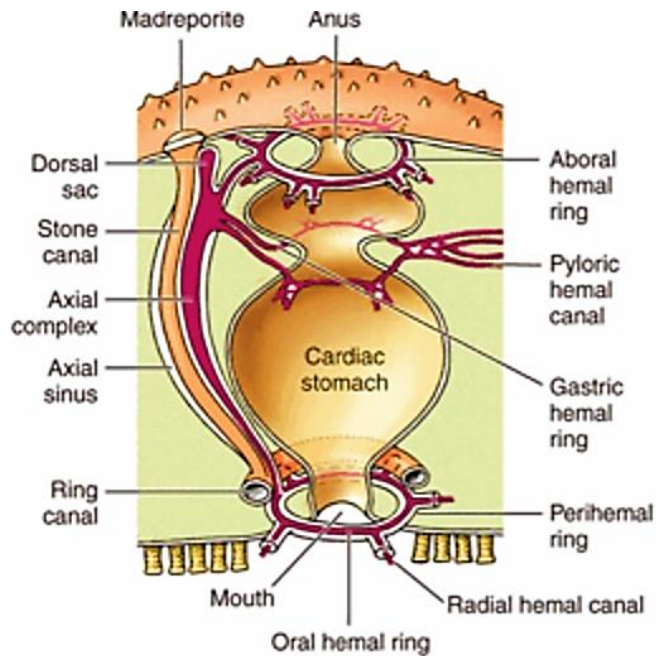
- They are strong deposit feeders
- Sea cucumbers have an interesting predator escape plan called **evisceration (内脏切除/吐脏)**: they expel the internal organs
- Since all echinoderms have regenerative capabilities, these internal organs will grow back



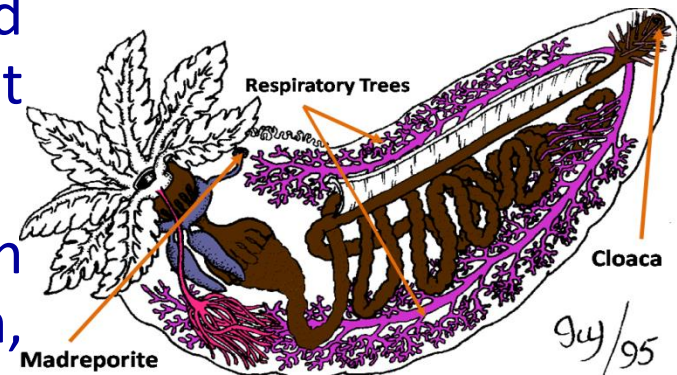
14.2 Biology of Echinoderms

- **Feeding and Digestion**

- ❑ The digestive system of echinoderms is relatively simple
- ❑ Many feed by extending, or everting, part of their stomach inside out through the mouth to envelop the food
- ❑ The stomach secretes digestive enzymes produced by large digestive glands that extend into the arms
- ❑ The digested food is carried into the glands for absorption and the stomach pulled back inside the body
- ❑ The gut of sea urchins and sea cucumbers is long and coiled
- ❑ Nutrients are transported in the fluid (coelomic fluid) that fills the extensive body cavity
- ❑ The coelomic fluid also transports oxygen because most echinoderms **lack a distinct circulatory system**



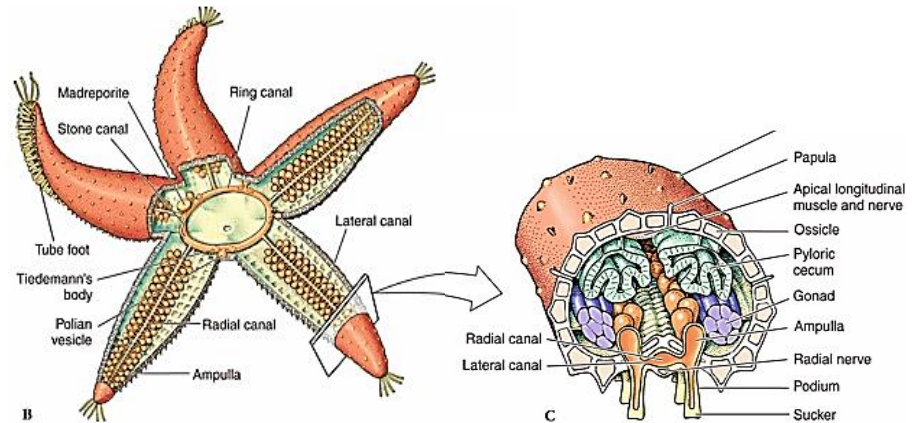
- In sea stars and sea urchins, gas exchange takes place across small, branched **projections of the body wall** connected at the base to the coelomic cavity
- In **sea cucumbers**, water is drawn in through the anus into a pair of thin, branched tubes called **respiratory trees**



9/4/95

- **Nervous System and Behavior**

- Knowledge is limited
- The nervous system coordinates movements of tube feet and spines in the absence of a brain
- More complex behaviors, such as the righting of the body after being turned over and camouflaging with bits of debris by sea urchins, are evidence that the nervous system may not be as simple as it looks



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Molecular mechanisms of tubulogenesis revealed in the sea star hydro-vascular organ

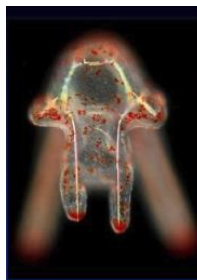
[Margherita Perillo](#) , [S. Zachary Swartz](#), [Cosmo Pieplow](#) & [Gary M. Wessel](#) 

[Nature Communications](#) **14**, Article number: 2402 (2023) | [Cite this article](#)

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- **Reproduction and Life History**

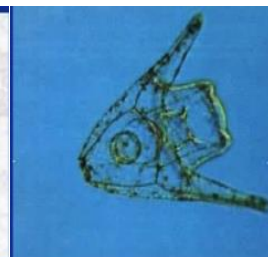
- ❑ The sexes are separate in most echinoderms
- ❑ In most group, 5, 10, or more gonads shed sperm or eggs directly into the water
- ❑ The gonads are usually located in the body cavity and open to the outside by way of a duct
- ❑ Gametes do not survive long in the water, so in many species individuals spawn all at once to ensure fertilization
- ❑ Development of the fertilized egg proceeds in the plankton and typically results in a ciliated larva characteristic of each group
- ❑ Echinoderm larvae are **bilaterally symmetrical** and it is not until metamorphosis that radial symmetry develops
- ❑ Some echinoderms **do not have planktonic larvae** and brood their offspring in special pouches or under the body



羽腕幼虫
(海星)



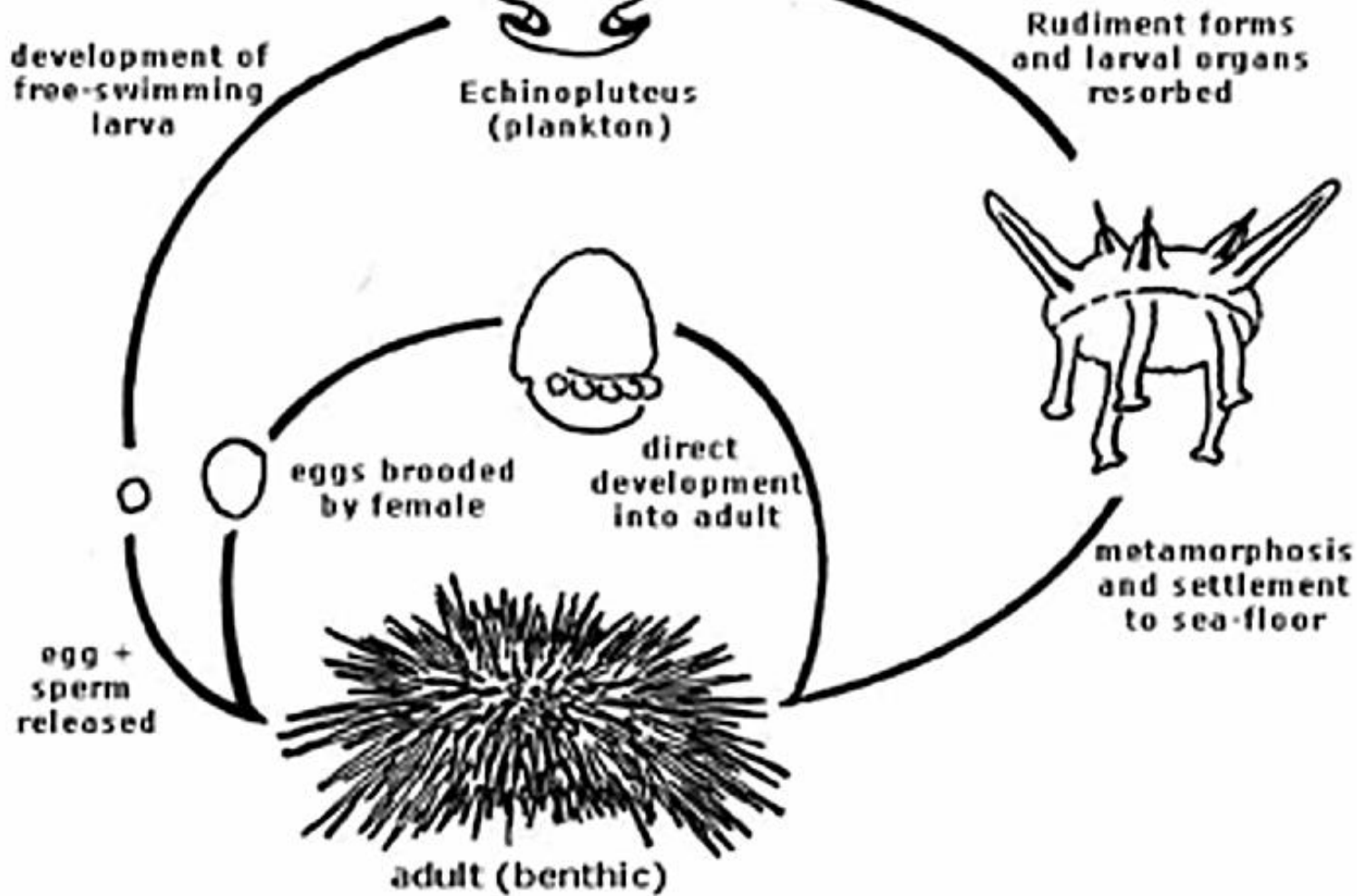
耳状幼虫
(海参)



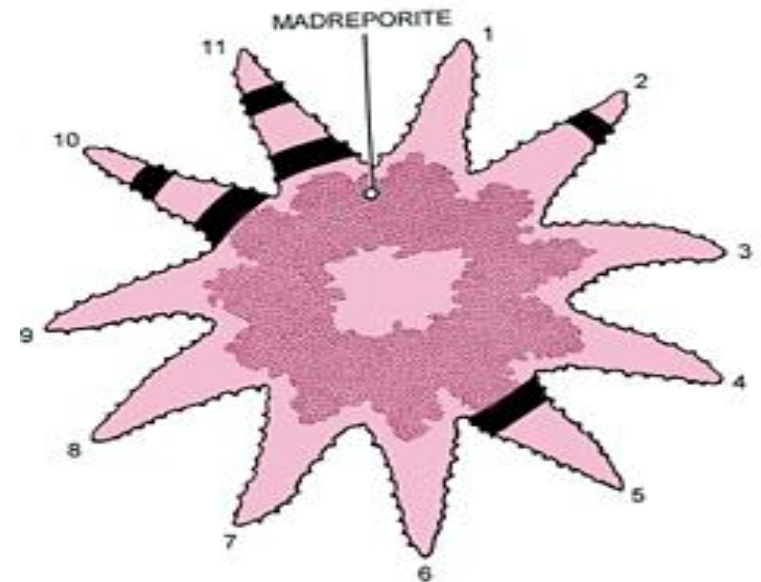
海胆幼虫
(海胆)



桶形幼虫
(海百合)

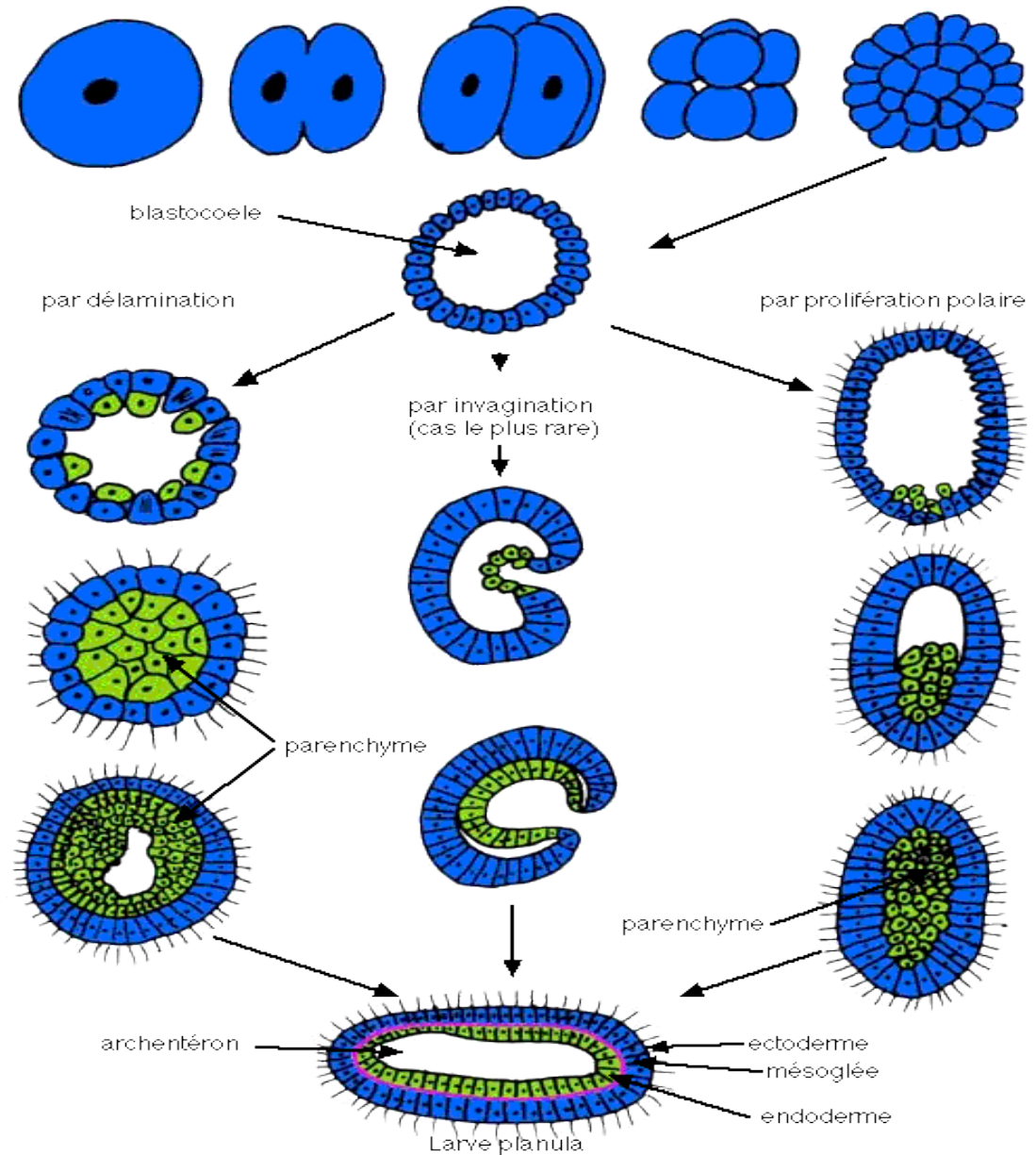


- ❑ Some sea stars, brittle stars, and sea cucumbers regularly reproduce asexually by the separation of the central disk or body into two pieces
- ❑ **Regeneration** is highly developed in echinoderms
- ❑ Sea stars, brittle stars, and sea lilies regenerate lost arms
- ❑ In some sea stars a severed arm can grow into a new individual
- ❑ In most sea stars only arms that include portions of the central disk can regenerate

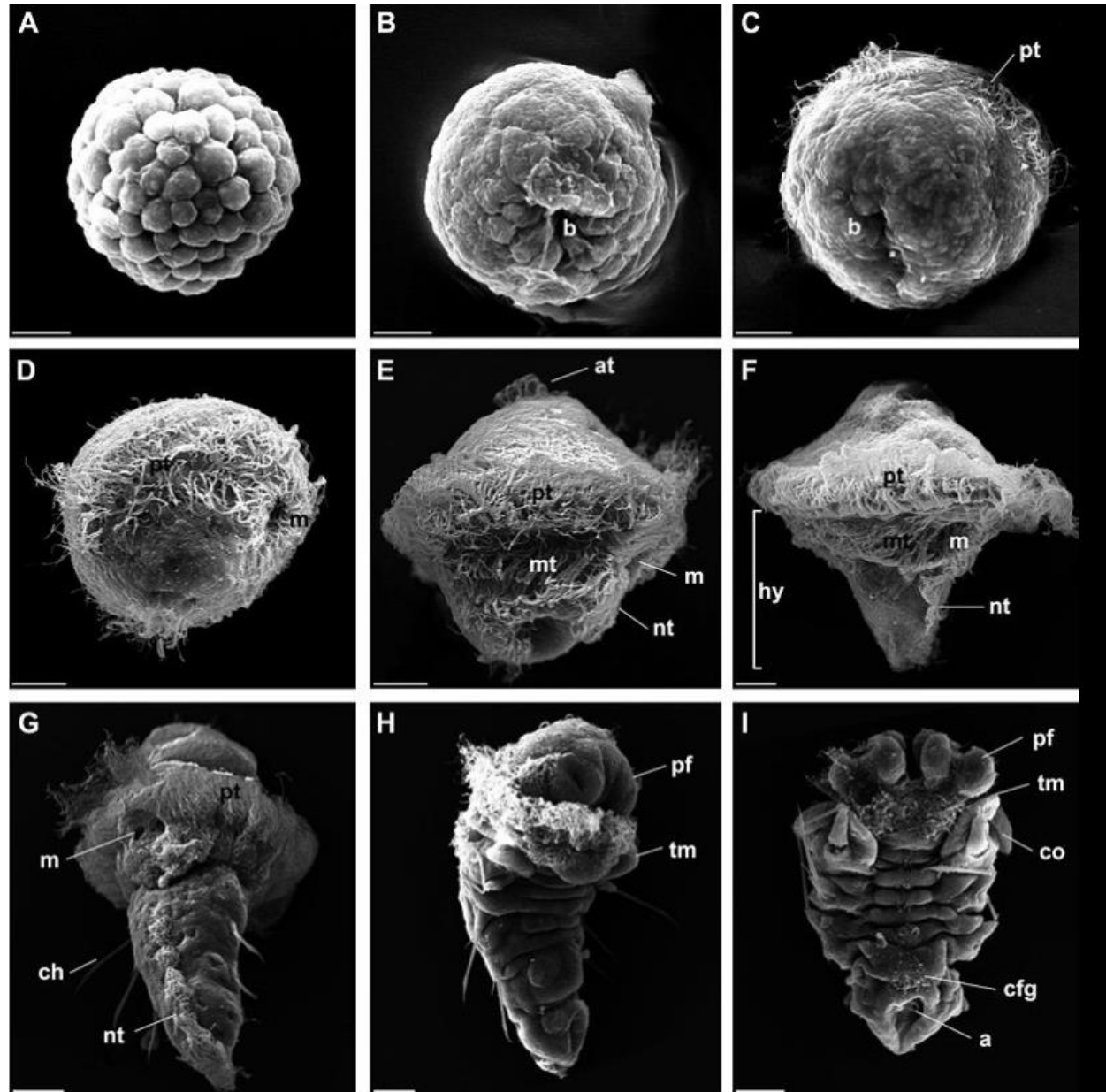


15. Invertebrate Larva

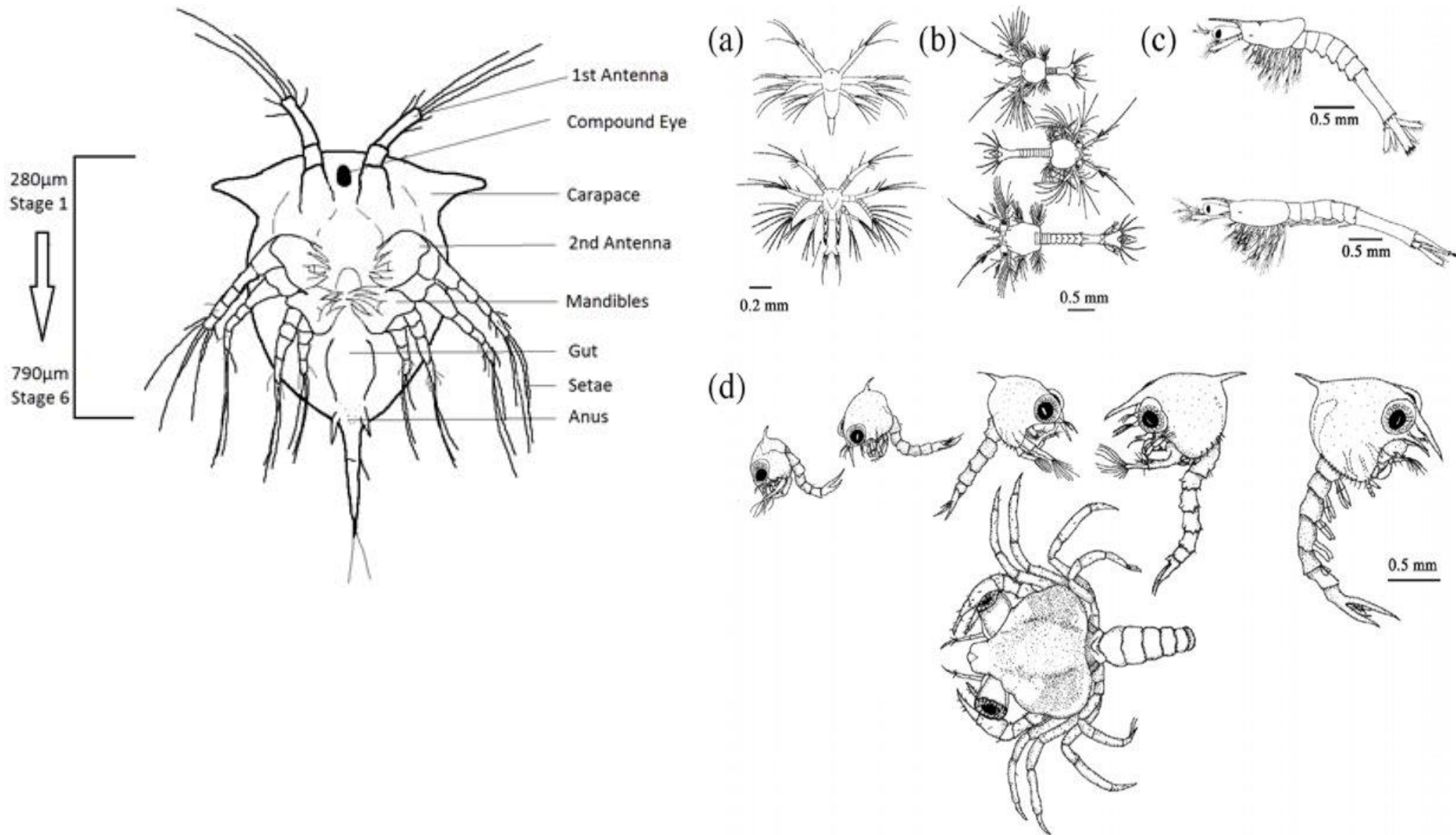
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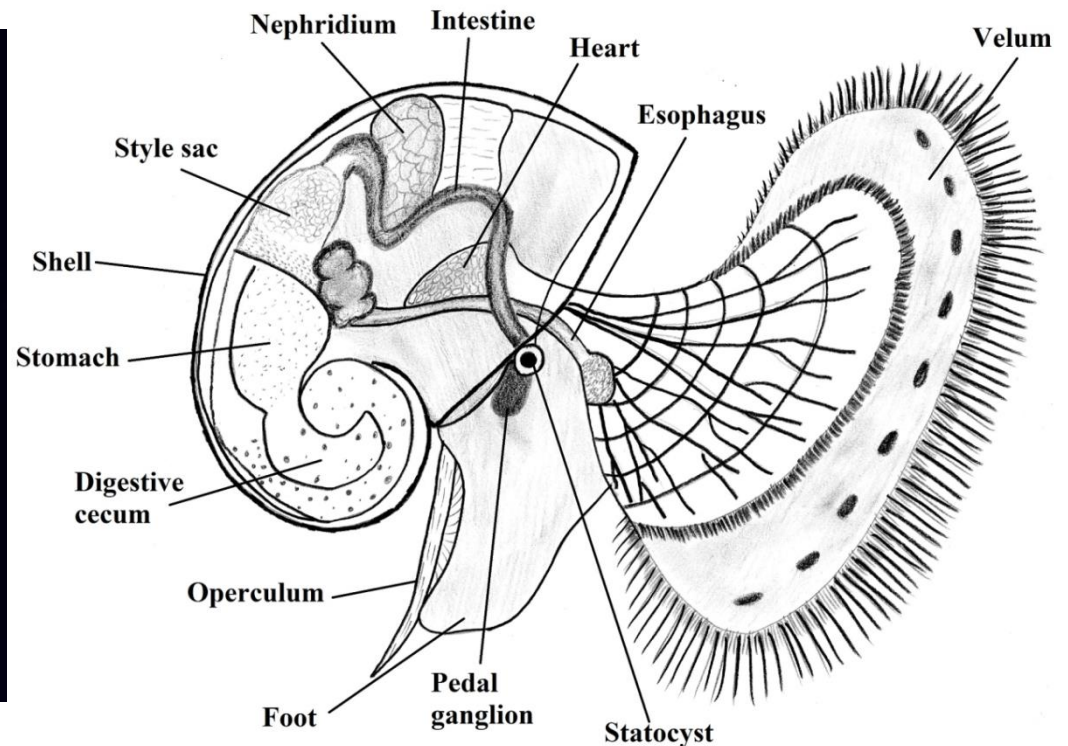
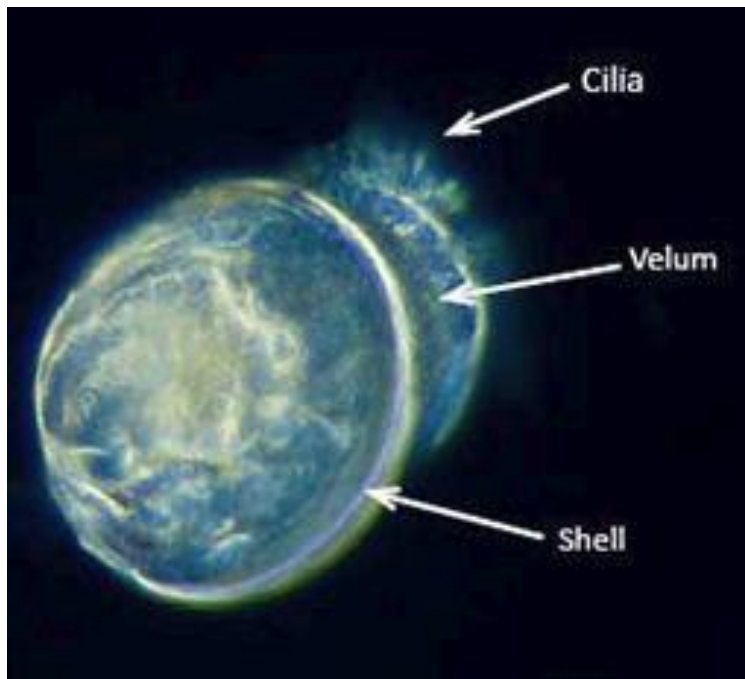
2. Trochophore:
exist as a larval
form including the
molluscs, annelids,
echiurans,
sipunculans and
nemerteans. It is
possible that
trochophore
larvae were
present in the life
cycle of the
group's common
ancestor

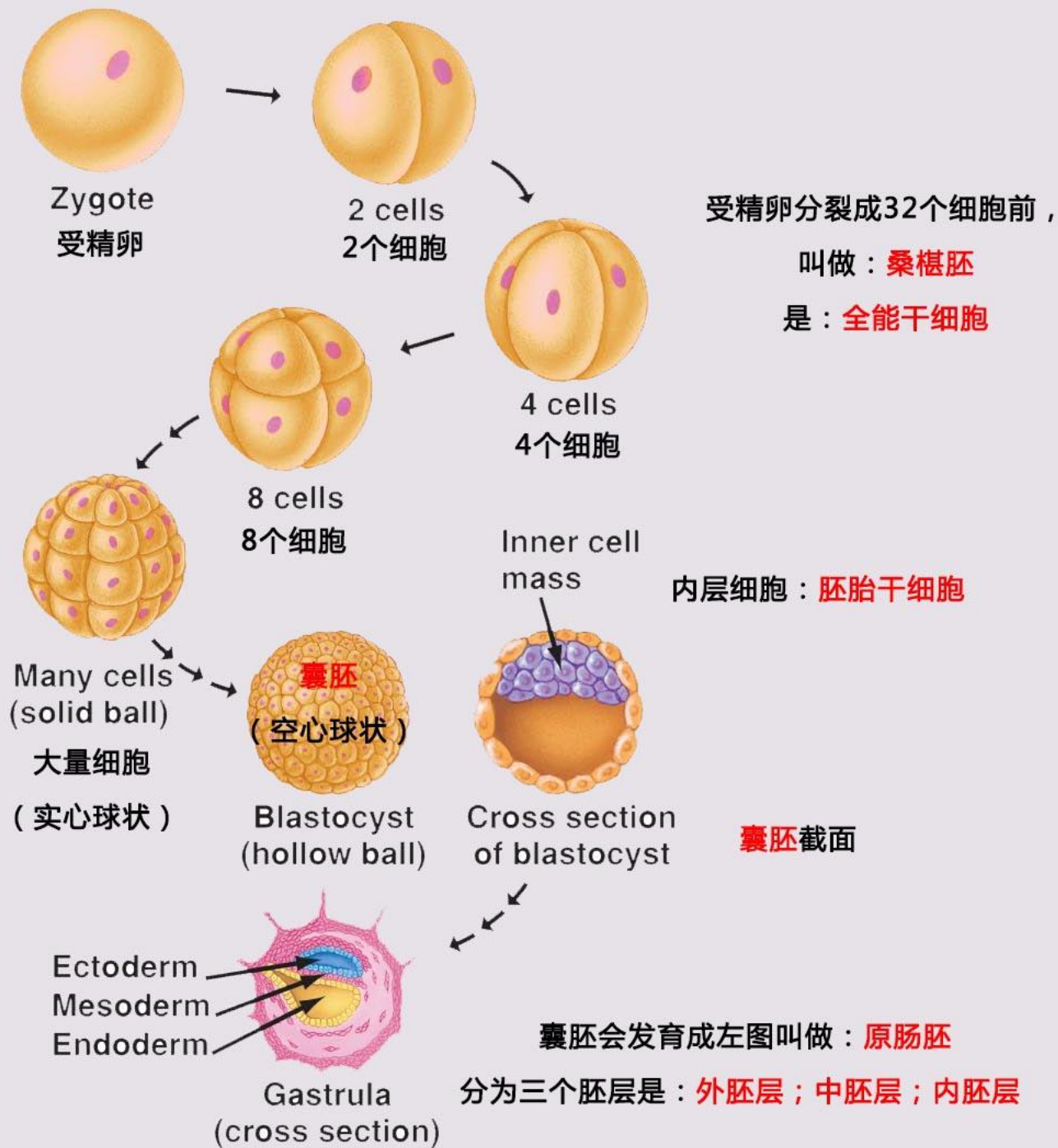


3. Nauplius: characterized by the use of the appendages of the head (the antennae) for swimming. The nauplius is also the stage at which a simple, unpaired eye first appears



4. Veliger: produced following either the embryonic or trochophore larval stage of development. The general structure of the veliger includes a shell that surrounds the visceral organs of the larva and a ciliated velum that extends beyond the shell as a single or multi-lobed structure used for swimming and particulate food collection





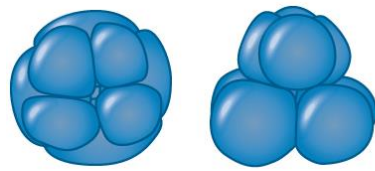
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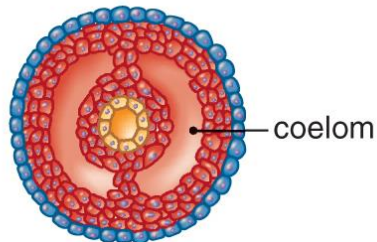
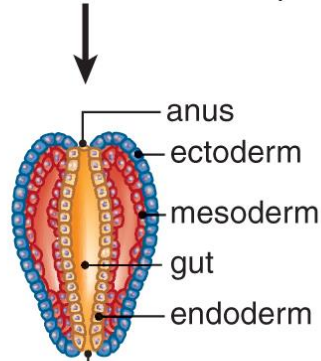
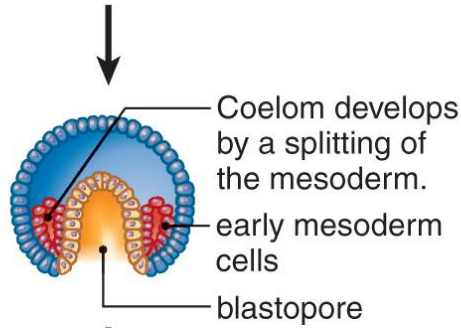
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Protostomia

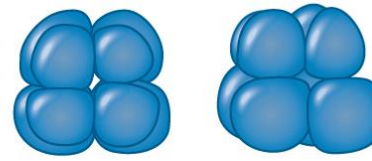


Top view Side view

Early cell division



Deuterostome



Top view Side view

Early cell division

